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Education

- April 2025 – Present ■ **Tokyo Metropolitan University (TMU), Ph.D.**
Major: Computer Science
Advisor: Prof. Dr. Masaki Aida
- Sep 2021 – Sept 2023 ■ **University of Engineering and Technology (UET), Lahore, M.Sc.**
Major: Computer Science
Advisor: Prof. Dr. Muhammad Shoaib
Thesis title: *A Technique To Enhance Social Media Reliance By Detecting Social Bots.*
- Oct 2017 – Aug 2021 ■ **University of Sargodha, BS.**
Major: Computer Science
Advisor: Prof. Muhammad Bilal
Thesis title: *Planting the Seeds of Sustainability with Plantobook.*

Research Publications

Peer-Reviewed Journal Articles

- **Owais, M.**, Nakajima, K., and Aida, M. (2026). "Detecting Evolving Social Bots Through Hybrid Linguistic and Profile-Based Feature Integration." *PLOS ONE*. (Under review)
- Shaikh, J. A., Wang, C., Sima, M. W. U., Arshad, M., **Owais, M.**, Hassan, D. S., et al. (2025). "A deep reinforcement learning-based robust intrusion detection system for securing IoMT healthcare networks." *Frontiers in Medicine*.
- Shaikh, J. A., Wang, C., Muhammad, W. U. S., Arshad, M., & **Owais, M.** (2024). "RCLNet: An Effective Anomaly-Based Intrusion Detection for Securing the IoMT System." *Frontiers in Digital Health*.
- **Owais, M.**, Shoaib, M., & Waseem, M. (2023). "A Holistic Approach for Detecting Socialbots on Twitter: Integration of Diverse Features." *The Nucleus*.

Conference Presentations

- **Owais, M.**, Nakajima, K., and Aida, M. (2025). "Text-Based Social Bot Detection Leveraging Linguistic Structures and Sentiment Features." In *Proceedings of the 14th International Conference on Complex Networks and their Applications*.
- Shaikh, J. A., Wang, C., **Owais, M.**, Zia, U., Sima, M. W. U., & Arshad, M. (2024, December). "FOID: A Feature-Optimized Intrusion Detection System for Securing IoMT Healthcare Networks." In *18th International Conference on Open Source Systems and Technologies (ICOSST)*, IEEE.

Working Papers and Work in Progress

- **Owais, M.**, Sajid, M., and Malik, U. H. (2026). "Self-Jailbreaking Large Language Models: Multi-Turn Adversarial Prompting and Cross-Model Guardrail Failures." Journal under preparation.

Professional Experience

Research Experience

- Apr 2025 – Present ■ **Masaki Aida Laboratory**, TMU, Tokyo
Research Assistant
Applied information network science and NLP-based social network analysis to model user behavior and network dynamics in large-scale systems. Analyzed phenomena such as echo chambers and online misinformation, developing data-driven insights and potential countermeasures. Utilized graph-based approaches, including spectral methods, to study complex network structures and information flow. Evaluated large-scale communication data to support network optimization and data-driven service strategies.
- Aug 2022 – Oct 2023 ■ **Center for Language Engineering (CLE)**, KICS, UET, Lahore
Research Officer
Contributed to multiple Urdu-focused AI projects, including automatic meeting minutes generation, speech separation, and multilingual speech corpus development. Developed intelligent speech-enabled systems for public service delivery (E-Khidmat) and worked on Urdu broadcast media content extraction and analytics, leveraging NLP, speech processing, and machine learning techniques.
- Mar 2022 – Aug 2022 ■ **Data Science Lab (DSL)**, KICS, UET, Lahore
Research Intern
Contributed to advanced data science research on large-scale and complex datasets, focusing on data collection, integration, and information extraction using machine learning and data mining techniques. Supported the development of scalable data processing systems to address real-world challenges involving high-volume, heterogeneous, and noisy data.

Teaching Experience

- Sep 2024 – Jan 2025 ■ **University of Engineering and Technology (UET), Lahore**
Course Instructor (Undergraduate)
Application to information and communication technology (1st Semester)
- Aug 2023 – Feb 2025 ■ **Superior University Lahore**, Computer Science
Course Designer and Instructor (Undergraduate)
Programming Fundamentals (1st Semester), Object-Oriented Programming (2nd Semester), Introduction to Logic (1st Semester), Data Structures and Algorithms (4th Semester)

Industry Experience

- Jul 2021 – Mar 2022 ■ **CodeFreaks**, Lahore
Junior React Native Developer
Designed and implemented smooth animations and innovative loading strategies to enhance user experience, ensuring a responsive and visually engaging interface. Optimized performance and usability by improving interaction flow, and delivering a polished, intuitive application experience.
- Jan 2021 – June 2021 ■ **LearnToEarn**, Mianwali
Junior Android Developer
Developed and maintained Android applications, implementing user-friendly interfaces and core functionalities. Collaborated with the team to debug issues, optimize performance, and enhance app usability.

Research Projects

- Oct 2022 – Oct 2024 ■ **Automatic Meeting Minutes and Analytics Generation for Urdu**
Center for Language Engineering (CLE), KICS, UET, Lahore
Designed and developed an intelligent automatic meeting minutes generation system for Punjabi, Urdu and English speakers, including speech separation models to handle overlapping audio. Built and curated a 15-hour multi-speaker meeting speech corpus to support speech processing and downstream applications.
- Jul 2021 – Aug 2023 ■ **Automated Urdu Broadcast Media Content Extraction and Analytics**
Center for Language Engineering (CLE), KICS, UET, Lahore
Developed a system to enable efficient extraction and management of spoken and written Urdu content from online digital videos. Implemented advanced content processing and intelligent analytics techniques for Urdu language data. Enhanced accessibility by enabling users to search and retrieve Urdu video content through an online search platform.
- Jan 2022 – Mar 2023 ■ **Expanding E-Khidmat Center's Services and Delivery to Broader Citizen Base Using Intelligent Urdu Language Technology**
Center for Language Engineering (CLE), KICS, UET, Lahore
Developed an intelligent, speech-enabled service delivery platform in local languages to improve accessibility and streamline access to public services. Evaluated user interactions and feedback to continuously enhance system usability, relevance, and overall service effectiveness.

Presentations, Talks, & Academic Services

- 2026 ■ Presented a research poster titled “**Decoding Non-Human Language: Detecting Bots and AI-Generated Text on Reddit**” at the **System Design Poster Presentation**, Tokyo Metropolitan University (TMU).
- 2025 ■ Participated as a judge for project evaluations at the **Information Technology Exhibition & Competition (ITEC)**, UET, an in-house talent hunt and technical event organized in collaboration with industrial partners.
- 2024 ■ Delivered a session at **Technical Education & Vocational Training Authority (TEVTA)** on Machine Learning & AI: Concepts and Applications, focusing on practical perspectives and real-world applications.
- Delivered a session at the **Technical Education & Vocational Training Authority (TEVTA)**, Lahore, on the importance of Python and its applications.
- Accompanied students from Superior University Lahore to CodeEx 2024, a Speed Programming Competition hosted at **Namal University, Mianwali**.

Mentorship

- *Hamad & Sana*, Department of Computer Science, Superior University Lahore (2023–2024). Supervised undergraduate research project titled “Advancing Urdu Speech Recognition through Dataset Construction and Fine-Tuning.”
- *Ahmed & Mujtaba*, Department of Computer Science, Superior University Lahore (2023–2024). Supervised undergraduate research project titled “Leveraging Phonetic Algorithms and Word Embeddings for Improved Roman Urdu Sentiment Analysis.”
- *Umair & Sidra*, Department of Computer Science, Superior University Lahore (2024). Supervised undergraduate research project titled “Action Item Identification Using Multimodal Meeting Transcripts.”

Awards and Achievements

- 2025 ■ **JSPS Research Funding.**
Awarded for doctoral research support in Japan.
- **Tokyo Global Partner (TGP) Scholarship.**
Fully funded scholarship awarded for Ph.D. studies at Tokyo Metropolitan University, Japan.
- 2023 ■ **Best Poster Presentation Award.**
Awarded by the Department of Computer Science, University of Engineering and Technology (UET), Lahore.
- 2022 ■ **Prime Minister's Youth Laptop Scheme.**
Awarded based on high academic merit in the department.
- 2021 ■ **Merit-Based Scholarship.**
Awarded during M.S. studies for outstanding academic performance.
- 2017 ■ **Fully Funded Undergraduate Scholarship.**
Awarded for four-year Bachelor's degree based on academic excellence.

Certification & Courses

- 2026 ■ **Transformers & Large Language Models**
Stanford University
- 2025 ■ **Gen AI Foundational Models for NLP & Language Understanding**
IBM
- **Deep Learning and Reinforcement Learning**
IBM
- **NLP: Twitter Sentiment Analysis**
Coursera
- 2024 ■ **Machine Learning**
Stanford University
- 2022 ■ **Python Core**
Sololearn

Skills

- Coding ■ Python / Flask / AI / ML / DL / NLP / PyTorch / TensorFlow / Transformers / PHP / WordPress / HTML / CSS / C / C++ / Redux / SQL / React Native
- Tools ■ Git / Microsoft Office / Kaldi Toolkit / Firebase / Shell (Bash) / $\text{\LaTeX}$ (Overleaf / R Markdown) / Ubuntu
- Misc. ■ Academic Research / Teaching / Training / Consultation

Research Interests

Computational and data-driven approaches for Social Network Analysis and Intelligent Systems, Natural Language Processing (NLP), Large Language Models (LLMs), LLM Security, Speech Recognition, Speaker Identification, and Multimodal Language Understanding. My research further explores Deep Learning (DL) and Machine Learning (ML) methodologies.

Languages

- English ■ Business
- Urdu ■ Native